Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

Нассер Мохамад

2026-03-17

Содержание (i)

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
nessermohammed@nessermohammed:~$ ls /etc/ > file.txt
nessermohammed@nessermohammed:~$ ls >> file.txt
nessermohammed@nessermohammed:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
```

Рисунок 1: Запись в файл

Работа фильтра

```
nessermohammed@nessermohammed:~$
nessermohammed@nessermohammed:~$ grep .conf file.txt > conf.txt
nessermohammed@nessermohammed:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/nessermohammed/.npm/_cacache/index-v5/84/ce
/home/nessermohammed/.npm/_cacache/index-v5/09/c0
/home/nessermohammed/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc2
9def30220242c53
/home/nessermohammed/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55
fb2000f59e2edcc
/home/nessermohammed/.npm/_cacache/index-v5/c9
/home/nessermohammed/.npm/_cacache/index-v5/cd
/home/nessermohammed/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350
482a1d990504757
/home/nessermohammed/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310
cda0b203217f693
/home/nessermohammed/.npm/_cacache/index-v5/c0
/home/nessermohammed/.npm/_cacache/index-v5/af/cf
/home/nessermohammed/.npm/_cacache/index-v5/af/3b/c01fad9ea9ac6256e90d5a59e2c8b816cc7bcf0f41a5c
a46eaa0bbab8743
/home/nessermohammed/.npm/_cacache/index-v5/c3
/home/nessermohammed/.npm/_cacache/index-v5/95/f7/cabf02718d2ced903e42baa9d89ccac492bdd331ae80e
550538ac79b0d3a
/home/nessermohammed/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f250621
4f94db8861160d3
/home/nessermohammed/.npm/_cacache/index-v5/ca
/home/nessermohammed/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188
a44d554f3f3ba2e
/home/nessermohammed/snap/hugo/common
/home/nessermohammed/snap/hugo/current
/home/nessermohammed/conf.txt
nessermohammed@nessermohammed:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: ‘/etc/libvirt’: Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: ‘/etc/lvm/archive’: Permission denied
/etc/logrotate.d/httpd
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
:find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
find: ‘/etc/userdb’: Permission denied
```

Рисунок 4: Поиск файлов

Управление процессами

```
nessermohammed@nessermohammed:~$
nessermohammed@nessermohammed:~$ find /etc -name "h*" -print | less
nessermohammed@nessermohammed:~$
nessermohammed@nessermohammed:~$
nessermohammed@nessermohammed:~$ find ~ -name "log*" > logfile &
[1] 8366
nessermohammed@nessermohammed:~$
[1]+  Done                    find ~ -name "log*" > logfile
nessermohammed@nessermohammed:~$ rm logfile
nessermohammed@nessermohammed:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
nessermohammed@nessermohammed:~$
nessermohammed@nessermohammed:~$ gedit &
[1] 8402
nessermohammed@nessermohammed:~$ ps | grep gedit
   8402 pts/3    00:00:00 gedit
nessermohammed@nessermohammed:~$ kill 8402
[1]+  Terminated              gedit
nessermohammed@nessermohammed:~$
nessermohammed@nessermohammed:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page  documents  the  GNU version of df.  df displays the amount of space
       available on the file system containing each file name argument.  If no  file  name  is
       given,  the  space  available on all currently mounted file systems is shown.  Space is
       shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is  set,
       in which case 512-byte blocks are used.

       If  an  argument  is  the absolute file name of a device node containing a mounted file
       system, df shows the space available on that file system rather than on the file system
       containing the device node.  This version of df cannot show the space available on  un-
       mounted  file  systems, because on most kinds of systems doing so requires non-portable
       intimate knowledge of file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all file  systems
       by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes  rather  than device usage; although the apparent size is
              usually smaller, it may be larger due to holes  in  ('sparse')  files,  internal
              fragmentation, indirect blocks, and the like

       -B, --block-size=SIZE
              scale  sizes  by SIZE before printing them; e.g., '-BM' prints sizes in units of
              1,048,576 bytes; see SIZE format below
```

Рисунок 8: Запуск команды df

Команды df и du

```
nessermohammed@nessermohammed:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89237944  65285720  58% /
devtmpfs         4011276        0   4011276   0% /dev
tmpfs            4035652      100   4035552   1% /dev/shm
tmpfs            1614264     1968   1612296   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035652   129812   4005840   1% /tmp
/dev/nvme0n1p3 155186176 89237944  65285720  58% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807128      192    806936   1% /run/user/1019
/dev/sr0         2677920  2677920         0 100% /run/media/nessermohammed/Fedora-WS-Live-43
nessermohammed@nessermohammed:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
115392  ./.npm
16      ./snap/hugo/25676/.config/go/telemetry/local
0       ./snap/hugo/25676/.config/go/telemetry/upload
16      ./snap/hugo/25676/.config/go/telemetry
16      ./snap/hugo/25676/.config/go
16      ./snap/hugo/25676/.config
16      ./snap/hugo/25676
0       ./snap/hugo/common
20      ./snap/hugo
20      ./snap
0       ./monthly
0       ./reports/monthly/monthly
0       ./reports/monthly
0       ./reports
4       ./ski.plases/equipment
0       ./ski.plases/plans
4       ./ski.plases
0       ./australia
0       ./play/games/play
0       ./play/games
0       ./play
1605204 .
nessermohammed@nessermohammed:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/nessermohammed/.npm/_cacache/index-v5/30/fa
/home/nessermohammed/snap
/home/nessermohammed/snap/hugo
/home/nessermohammed/snap/hugo/25676
/home/nessermohammed/snap/hugo/25676/.config
/home/nessermohammed/snap/hugo/25676/.config/go
/home/nessermohammed/snap/hugo/25676/.config/go/telemetry
/home/nessermohammed/snap/hugo/25676/.config/go/telemetry/local
/home/nessermohammed/snap/hugo/25676/.config/go/telemetry/upload
/home/nessermohammed/snap/hugo/common
/home/nessermohammed/monthly
/home/nessermohammed/reports
/home/nessermohammed/reports/monthly
/home/nessermohammed/reports/monthly/monthly
/home/nessermohammed/ski.plases
/home/nessermohammed/ski.plases/equipment
/home/nessermohammed/ski.plases/plans
/home/nessermohammed/australia
/home/nessermohammed/play
/home/nessermohammed/play/games
/home/nessermohammed/play/games/play
nessermohammed@nessermohammed:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.